

Analysis Of UWA Communication on Dependent Parameters Using Bellhop Simulator

Abstract:

Simulation forms an important part of the development and empirical evaluation of underwater acoustic network (UAN) protocols. The key feature of a credible network simulation model is a realistic channel model. A common approach to simulating realistic underwater acoustic (UWA) channels is by using specialised beam tracing software such as BELLHOP. However, BELLHOP and similar modeling software typically require knowledge of ocean acoustics and a substantial programming effort from UAN protocol designers to integrate it into their research. This paper is a distilled tutorial on UWA channel modeling with a focus on network simulation, providing a trade-off between the flexibility of low level channel modeling via beam tracing and the convenience of automated channel modeling, e.g. via the World Ocean Simulation System (WOSS). The tutorial is accompanied by our MATLAB simulation code that interfaces with BELLHOP to produce channel data for UAN simulations. As part of the tutorial, we describe two methods of incorporating such channel data into network simulations, including a case study for each of them: 1) directly importing the data as a look-up table, 2) using the data to create a statistical channel model. The primary aim of this paper is to provide a useful learning resource and modeling tool for UAN protocol researchers. Initial insights into the UAN protocol design and performance provided by the statistical channel modeling approach presented in this paper demonstrate its potential as a powerful modeling tool for future UAN research.

Keywords: Underwater Communication, UWAC, BELLHOP.

1. INTRODUCTION

Around 70% of the surface of earth is covered with water. It contains large amount of energy and natural resources and hence for exploring these resources, development of effective underwater wireless communication system became very important. Due to number of applications like scientific marine exploring, pollution controlling, voice and data communications between divers, mine reconnaissance, study of disasters, military projects, etc. the field of underwater wireless communication is rapidly increasing. Also, shipbuilding and offshore engineering is taking lot of interest in it. So, obviously it is one of the important communication media for communication researchers.

Underwater wireless communication is an enabling technology for oceanography and its applications. It consists of number of vehicles with attached sensors for monitoring the given area.

But, development of underwater wireless communication system is not that much easy as it is totally different from terrestrial radio environment both in terms of energy, cost and channel propagation.

The basic problems in front of underwater acoustic communication are limited sound velocity, absorption of a sound in water, multipath propagation, ambient noise, Doppler's spread, etc.

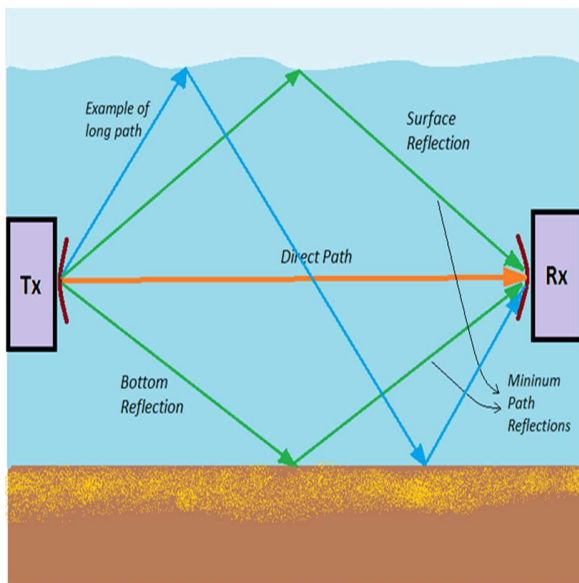
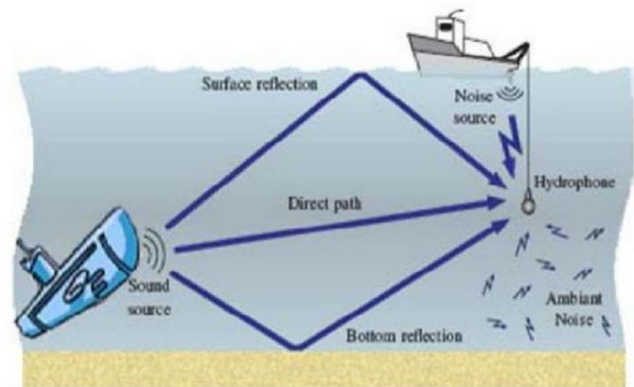


Fig 1: Overview of Underwater wireless communication

Underwater wireless acoustic communication system is a two way communication system in which digital data is converted into special underwater sound signals. Then these signals are received by another trans receiver and then converted back to digital data. This communication is done in multipath propagation of acoustic/sound signal.

Underwater wireless communications play an important role in marine activities such as environmental monitoring, underwater exploration, and scientific data collection. Underwater wireless communications still remain quite challenging, due to the unique and harsh conditions that characterize underwater channels. These conditions include for example severe attenuation, multipath dispersion, and limited resource utilization. Nonetheless advanced communication techniques using acoustic, electromagnetic and/or optical waves have

emerged to tackle fundamental and practical challenges of underwater wireless communication.



Underwater acoustic communication is a technique of sending and receiving messages below water. There are several ways of employing such communication but the most common is by using hydrophones. Underwater communication is difficult due to factors such as multi-path propagation, time variations of the channel, small available bandwidth and strong signal attenuation, especially over long ranges. Compared to terrestrial communication, underwater communication has low data rates because it uses acoustic waves instead of electromagnetic waves.

2. PROPOSAL METHOD

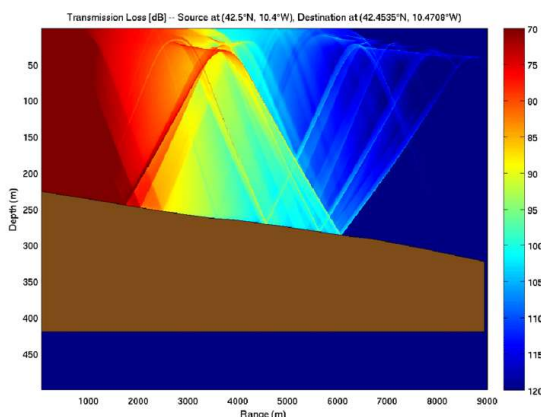
Underwater acoustic communication is an essential part of many applications such as underwater sensing, oceanography, and defense operations. The performance of underwater acoustic communication systems is highly dependent on various parameters such as transmission range, frequency, depth.

TRANSMISSION RANGE

Transmission range is one of the most critical parameters in underwater acoustic (UWA) communication systems. It refers to the distance that an acoustic signal can travel between the transmitter and receiver without significant loss or degradation of the signal. The transmission range is affected by various factors such as the frequency of the signal, the depth of the water, and the properties of the medium through which the signal travels.

TRANSMISSION LOSS

Transmission loss is a phenomenon that occurs when acoustic signals are transmitted through water in underwater acoustic (UWA) communication systems. It refers to the decrease in signal power or intensity as the signal travels through water. The transmission loss is caused by various factors such as absorption, scattering, and reflection of sound waves as they propagate through water. The attenuation of sound waves due to absorption



is one of the primary causes of transmission loss in UWA communication systems.

FREQUENCY

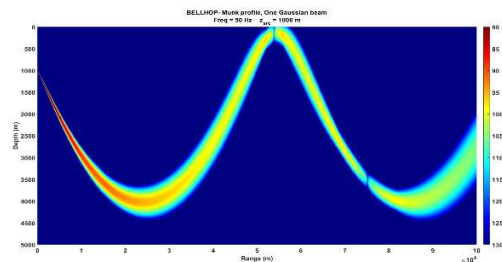
Frequency is a crucial parameter in underwater acoustic (UWA) communication

systems as it affects the signal propagation and transmission characteristics in water. The choice of frequency depends on several factors, such as the desired range of communication, the type of application, and the environmental conditions in which the system will operate.

Low-frequency signals, such as those in the range of 10-100 kHz, can propagate long distances in water, making them suitable for long-range communication.

DEPTH

Depth is an important factor to consider in underwater acoustic (UWA) communication systems as it affects the propagation

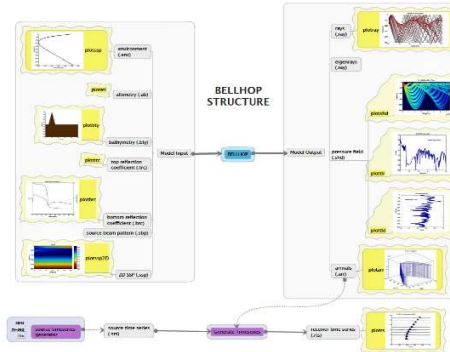


characteristics of acoustic signals in water. As depth increases, the water pressure and density increase, leading to changes in the speed of sound, attenuation, and scattering of acoustic signals.

3.BELLHOP

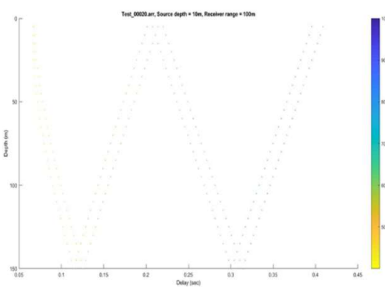
BELLHOP is a beam tracing model for predicting acoustic pressure fields in ocean environments. BELLHOP can produce a variety of useful outputs including transmission loss, eigenrays, arrivals, and received time-series. It also allows for range-dependence in the top and bottom boundaries. BELLHOP is implemented in Fortran, Matlab,

and Python and used on multiple platforms (Mac, Windows, and Linux).

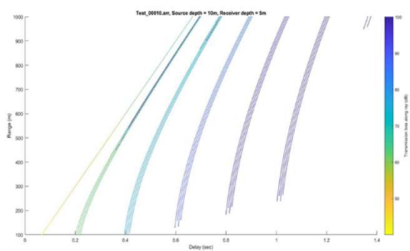


4.results

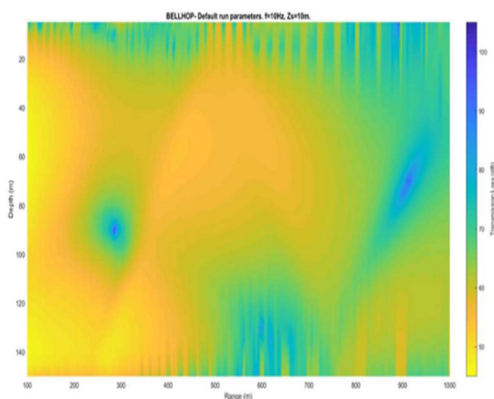
Delay vs depth



Delay vs Range



Range Vs Depth



5.CONCLUSION

In conclusion, underwater acoustic communication is a complex process that dependent on several parameters such as transmission loss, range, depth, and multiple frequencies. The Bellhop ray-tracing software is a useful tool for predicting the transmission loss of sound waves at different frequencies over a given range in underwater environments. The software takes into account factors such as the frequency of the sound waves, the properties of the medium, and the presence of obstacles, to provide accurate predictions of transmission loss in different scenarios.

REFERENCE

1. J. Peterson and M. Porter, "Ray/Beam Tracing for Modelling the Effects of Ocean and Platform Dynamics", IEEE Journal of Oceanic Engineering, vol. 38, no. 4, pp. 655-665, 2013.
2. Porter M.B. and Bucker H.P. Gaussian beam tracing for computing ocean acoustic fields. J. Acoust. Soc. America, 82(4):1349-1359,1987.
3. Jensen F., Kuperann W., Porter M., and Schmidt H. Computational Ocean Acoustics. AIP Series in Modern Acoustics and Signal Processing, New York, 1994.
4. MacGillivray, "Modelling underwater sound propagation from an airgun array using the parabolic equation method", The Journal of the Acoustical Society of America, vol. 122, no. 5, p. 2942,2007.

5. N. Chotiros and M. Stern, "High-frequency acoustic wave propagation modelling in shallow water", The Journal of the Acoustical Society of America, vol. 100, no. 4, p. 2701, 1996.
6. R. Frosch, "Underwater Sound: Deep-Ocean Propagation: Variations of temperature and pressure have great influence on the propagation of sound in the ocean.", Science, vol. 146, no. 3646, pp. 889-894, 1964.
7. Milica Stojanovic, "Underwater Acoustic Communication Channels: Propagation Models and Statistical Characterization. ", IEEE Communications Magazine, January 2009.

M. B. Porter, Ocean Acoustics Library, Bellhop,
<http://oalib.hlsresearch.com/>

.